Notes: Besley and Burgess (QJE, 2002)

The Political Economy of Government Responsiveness: Theory and Evidence from India

Contents

1 Big picture

- **Question.** What makes democratic governments responsive to the needs of vulnerable citizens?
- **Setting.** The paper studies Indian state governments from 1958 to 1992, focusing on responses to food production shocks and flood damage.
- **Main theory.** Government responsiveness is a political agency problem. Opportunistic incumbents exert effort when voters can observe and reward that effort.
- **Key mechanism.** Mass media improves information flows. If vulnerable citizens learn that incumbents respond to shocks, they are more likely to support those incumbents.
- **Political accountability channel.** Responsiveness should be stronger where turnout is higher, political competition is stronger, the vulnerable population is larger, and media access is greater.
- **Main empirical outcomes.** The paper studies two policy responses: public food distribution and calamity relief expenditure.
- **Main shocks.** Need is measured by food grain production shocks and flood damage, both driven by rainfall extremes.
- **Central empirical finding.** Newspaper circulation is strongly associated with higher government activism and with stronger policy responses to shocks.
- **Political result.** Public food distribution is higher when political competition is stronger and around election years. Calamity relief responds less to these purely political variables.
- **IV result.** Instrumenting newspaper circulation using newspaper ownership structure preserves the positive role of newspapers in responsiveness.
- **Takeaway.** Democratic institutions work better for vulnerable populations when citizens have information. Media development and political competition increase the electoral payoff to responsive governance.

2 Incremental contribution of the paper

2.1 Contribution to political economy

- Earlier political economy work emphasizes electoral competition, voting, and redistribution, but often treats voter information as implicit.
- Besley and Burgess explicitly model government responsiveness as a political agency problem in which voters use observed relief effort to infer incumbent type.
- The paper links three objects in one framework: information, electoral incentives, and public policy responses to shocks.

Interpretation: The incremental contribution is to show that democratic accountability is not only about elections. Accountability requires voters to know what governments do.

2.2 Contribution to development economics

- Much development work focuses on poverty, shocks, and social protection, but less often identifies political mechanisms behind public relief.
- This paper shows that state capacity in social protection varies systematically with media penetration and electoral incentives.

- It studies actual public relief systems used by vulnerable populations, rather than only program participation or household outcomes.

Interpretation: The paper contributes a political explanation for why similarly poor regions may receive different levels of protection from governments.

2.3 Contribution to the economics of media

- The paper is one of the early empirical economics papers linking mass media to policy responsiveness.
- Rather than studying media bias or media ownership alone, it studies media as an accountability technology.
- It uses both levels of newspaper circulation and interactions between newspaper circulation and shocks.

Interpretation: The interaction tests are especially important: newspapers do not just correlate with higher spending; they make spending more responsive to objective needs.

2.4 Contribution to empirical public finance and governance

- The paper combines a formal agency model with state-level panel data over more than three decades.
- It uses state fixed effects, year fixed effects, economic controls, political controls, and an instrumental variables strategy.
- It distinguishes chronic activism from shock responsiveness, which makes the empirical interpretation more precise.

Interpretation: The study is an early example of using subnational panel variation to study political accountability and government performance in a developing country.

3 Theory: political agency and responsiveness

3.1 Basic actors

The model has two periods and two types of citizens:

- **Vulnerable citizens,** who may need public action such as food distribution or disaster relief.
- **Nonvulnerable citizens,** who vote mainly on ideological grounds.

There are two possible incumbent types:

- **Altruistic incumbents,** who put in effort to help vulnerable citizens.
- **Opportunistic incumbents,** who exert effort only when it helps reelection.

Interpretation: The model is designed to explain why even selfish incumbents may respond to citizen needs when information and voting incentives are strong enough.

3.2 Media and voter information

Let e denote incumbent effort and m denote media access. If an incumbent exerts effort in period 1, the share of vulnerable citizens who learn about this effort is

$$s(e, m, \rho) = p(e, m)\rho + (1 - \rho)q(e, m), \tag{1}$$

where ρ is the fraction of vulnerable citizens who experience the shock directly.

- $p(e, m)$ is the probability that citizens who directly experience the shock learn about effort.
- $q(e, m)$ is the probability that citizens who do not directly experience the shock learn about effort.
- Both probabilities increase with effort and media access.

Interpretation: Media access matters because it converts hidden government effort into politically valuable information.

3.3 Election probability

The incumbent wins when informed vulnerable citizens plus ideologically supportive nonvulnerable citizens form a majority. The probability of winning can be written as

$$P(e; m, \rho, a, b, \gamma) = \begin{cases} 1, & \gamma a s > \frac{1}{2} - (1-\gamma)a, \\ \dfrac{(2b-a) + \gamma a(1-\gamma)s - \frac{1}{2}(1-\gamma)}{2(b-a)}, & \text{interior case,} \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Here:

- γ is the vulnerable population share;
- a is turnout among vulnerable voters;
- b measures expected ideological support for the incumbent;
- the election is more competitive when the incumbent has less advantage.

Interpretation: The incumbent's incentive to respond is strongest when effort can swing the election.

3.4 Effort choice

An opportunistic incumbent chooses effort to maximize expected reelection benefits net of effort cost:

$$\max_e \{P(e; m, \rho, a, b, \gamma)\beta - e\}. \tag{3}$$

In an interior solution, the first-order condition is

$$\frac{\gamma a}{2(b-a)(1-\gamma)} \left[p_e(e^*, m)\rho + (1-\rho)q_e(e^*, m)\right]\beta = 1. \tag{4}$$

Interpretation: Media, turnout, political competition, and the size of the vulnerable population all raise the marginal electoral benefit of effort.

3.5 Main predictions

The model predicts that government effort should be higher when:

1. media access is higher;
2. turnout is higher;
3. the vulnerable population share is larger;
4. the incumbent has a lower political advantage, meaning competition is stronger;
5. shocks are more visible to the people affected;
6. elections are closer, if voter memory or campaign salience matters.

Interpretation: The empirical analysis tests these predictions using Indian states, natural shocks, newspapers, and political competition.

4 Data and measurement

4.1 Panel structure

- The sample covers the sixteen major Indian states over 1958–1992.
- Haryana split from Punjab in 1965; from that year onward Punjab and Haryana enter separately.
- The maximum panel size is 552 state-year observations, with fewer observations in some regressions due to missing data.
- The main empirical design includes state fixed effects and year fixed effects.

Interpretation: State fixed effects absorb persistent differences across states, while year fixed effects absorb national shocks and national policy shifts.

4.2 Policy response variables

The two policy outcomes are:

- **Public food distribution per capita:** the amount of food distributed through India's public distribution system.
- **Calamity relief expenditure per capita:** state relief spending in response to natural calamities such as floods and droughts.

Interpretation: Public food distribution is more visible and potentially more politicized. Calamity relief is a more direct disaster-response expenditure category.

4.3 Need and shock variables

The main need variables are:

- **Food grain production per capita:** lower production proxies for food insecurity.
- **Flood damage per capita:** crop damage due to floods, measured in real per capita terms.
- **Drought and flood dummies:** rainfall extremes two standard deviations below or above state-specific rainfall means.

Interpretation: Rainfall variables provide exogenous climatic variation that helps interpret food production and flood damage as shocks rather than endogenous policy outcomes.

4.4 Media variables

- Newspaper circulation per capita is the primary media-development measure.
- The paper also separates circulation by language: English, Hindi, and other-language newspapers.
- Newspaper ownership variables are used as instruments in the IV analysis.

Interpretation: Language-specific circulation matters because local-language newspapers may reach vulnerable citizens more directly than English-language newspapers.

4.5 Political variables

- Turnout is lagged one election period.
- Political competition is measured as minus the absolute difference in legislative seat share between the dominant party and its main competitor.
- Election timing is captured by a dummy for election and pre-election years.

Interpretation: These variables proxy for how electorally valuable responsiveness is to the incumbent.

5 Empirical specifications

5.1 Baseline government activism regression

A baseline specification is

$$Y_{st} = \alpha_s + \lambda_t + \beta Shock_{st} + \delta Media_{st} + \theta Political_{st} + \Gamma X_{st} + \varepsilon_{st}, \tag{5}$$

where Y_{st} is either public food distribution or calamity relief expenditure.

- α_s are state fixed effects.
- λ_t are year fixed effects.
- X_{st} includes economic controls such as income, urbanization, population density, total population, and central transfers.

Interpretation: This regression asks whether states with greater media access and stronger accountability have higher levels of relief activity.

5.2 Responsiveness regression

The core responsiveness specification interacts shocks with newspaper circulation:

$$Y_{st} = \alpha_s + \lambda_t + \beta Shock_{st} + \delta Media_{st} + \kappa(Shock_{st} \times Media_{st}) + \Gamma X_{st} + \theta Political_{st} + \varepsilon_{st}. \tag{6}$$

- For public food distribution, the key shock is food grain production.
- For calamity relief, the key shock is flood damage.
- A significant interaction means governments respond more to shocks when media access is higher.

Interpretation: This is the most important empirical test of the theory: media should raise the responsiveness of policy to need, not merely overall spending levels.

5.3 Instrumental variables specification

The paper instruments newspaper circulation with ownership structure:

$$Newspaper_{st} = \Pi Ownership_{st} + \Gamma X_{st} + \alpha_s + \lambda_t + u_{st}. \tag{7}$$

The second stage replaces newspaper circulation and its interaction with predicted values.

Interpretation: Ownership variables proxy for the independence and competitiveness of the press. The IV strategy addresses concerns that newspaper circulation is endogenous to government performance or unobserved state development.

5.4 Political-interaction specification

The paper also tests whether responsiveness varies with political variables:

$$Y_{st} = \alpha_s + \lambda_t + \beta Shock_{st} + \delta Politics_{st} + \kappa(Shock_{st} \times Politics_{st}) + \Gamma X_{st} + \varepsilon_{st}. \tag{8}$$

Interpretation: The political interaction tests whether elections, turnout, and competition change the way governments respond to shocks.

6 Identification and empirical credibility

6.1 Strengths of the design

- The analysis uses a long state-year panel with fixed effects.
- The main shocks are plausibly exogenous climatic events or agricultural outcomes strongly driven by rainfall.
- The core test uses interactions between shocks and media, making it less likely that results are driven only by broad development differences.
- The IV approach uses newspaper ownership to address media endogeneity.

Interpretation: The strongest evidence is that policy becomes more responsive to objective shocks when media circulation is higher.

6.2 Threats to identification

- Newspaper circulation may proxy for literacy, civic engagement, or broader social development.
- Relief spending may be measured with error.
- Political variables may be endogenous to state-level political and economic changes.
- State governments may differ in administrative capacity in ways not fully captured by controls.

Interpretation: The paper's robustness comes from state fixed effects, economic controls, interaction designs, and IV estimation, but it remains an observational political economy study.

6.3 How to interpret causality

The most careful interpretation is:

> Higher newspaper circulation and political accountability are associated with greater responsiveness to shocks, consistent with the model that information improves democratic accountability.

Interpretation: The results are not a randomized experiment on media. They are strong panel evidence that informational institutions improve the working of democracy.

7 Tables and figures

7.1 Table I: summary statistics

Read:

- Table I reports state-level means and standard deviations for public food distribution, calamity relief, food grain production, flood damage, newspaper circulation, electoral turnout, political competition, and state income.
- Newspaper circulation differs sharply across states, from very low levels in Bihar and Madhya Pradesh to much higher levels in Kerala and Maharashtra.
- Calamity relief and flood damage are also highly heterogeneous across states.

Economic message: The table shows the cross-state variation that powers the analysis. India's federal structure provides wide variation in shocks, media development, and politics.

TABLE I
SUMMARY OF MAIN VARIABLES

State	Public food distribution	Calamity relief expenditure	Food grain production	Flood damage	Newspaper circulation	Other newspaper circulation	English newspaper circulation	Hindi newspaper circulation	Electoral turnout	Political competitiveness	State income
Andhra	11.615	3.941	185.851	8.252	0.029	0.0299	0.003	0.001	68.719	−0.558	1004
Pradesh	(9.012)	(3.399)	(18.448)	(14.937)	(0.011)	(0.012)	(0.001)	(0.001)	(3.515)	(0.113)	(260)
Assam	24.681	3.419	150.402	10.802	0.0186	0.0135	0.003	0.001	62.978	−0.552	903
	(7.632)	(3.124)	(12.973)	(11.729)	(0.009)	(0.007)	(0.002)	(0.001)	(11.530)	(0.241)	(196)
Bihar	11.110	1.491	141.008	6.724	0.020	0.003	0.002	0.014	51.764	−0.454	633
	(6.001)	(1.216)	(19.695)	(9.177)	(0.012)	(0.002)	(0.001)	(0.011)	(5.903)	(0.136)	(110)
Gujarat	18.576	5.414	118.376	3.599	0.054	0.053	0.002	0.0005	55.906	−0.568	1176
	(10.512)	(4.677)	(30.598)	(6.285)	(0.008)	(0.009)	(0.001)	(0.0004)	(5.678)	(0.253)	(272)
Haryana	9.813	2.840	467.687	8.799	0.020	0.004	0.004	0.013	67.431	−0.541	1444
	(4.081)	(2.102)	(99.335)	(15.280)	(0.005)	(0.002)	(0.004)	(0.005)	(5.108)	(0.237)	(357)
Jammu &	42.690	3.585	191.525	3.871	0.026	0.022	0.004	0.001	68.964	−0.547	1021
Kashmir	(11.219)	(5.629)	(30.503)	(12.672)	(0.010)	(0.006)	(0.003)	(0.001)	(5.533)	(0.280)	(228)
Karnataka	15.368	1.663	180.081	0.485	0.047	0.045	0.008	0.001	63.372	−0.587	1037
	(7.774)	(2.212)	(24.588)	(1.844)	(0.014)	(0.012)	(0.002)	(0.001)	(5.825)	(0.216)	(216)
Kerala	45.979	1.662	54.886	3.607	0.151	0.162	0.004	0.001	77.572	−0.152	864
	(19.337)	(3.441)	(10.324)	(7.715)	(0.060)	(0.064)	(0.003)	(0.001)	(3.772)	(0.123)	(182)
Madhya	7.564	1.383	255.743	0.552	0.0225	0.0004	0.001	0.020	49.089	−0.531	843
Pradesh	(5.333)	(1.802)	(31.733)	(2.300)	(0.017)	(0.002)	(0.002)	(0.017)	(6.056)	(0.145)	(190)
Maharashtra	28.271	2.752	147.700	0.339	0.117	0.055	0.0480	0.016	59.347	−0.674	1288
	(8.617)	(5.352)	(29.260)	(0.695)	(0.017)	(0.007)	(0.015)	(0.008)	(4.384)	(0.183)	(331)

Table I, part 1: state-level summary statistics

Orissa	10.944	4.673	222.052	5.604	0.016	0.018	0.001	0.0004	44.939	−0.413	873
	(5.082)	(5.625)	(31.243)	(8.093)	(0.010)	(0.011)	(0.0005)	(0.0005)	(7.490)	(0.255)	(186)
Punjab	15.952	4.978	668.551	9.946	0.058	0.045	0.004	0.012	66.139	−0.384	1732
	(12.328)	(8.058)	(206.580)	(19.041)	(0.019)	(0.014)	(0.003)	(0.007)	(4.077)	(0.223)	(384)
Rajasthan	10.209	5.000	229.405	2.188	0.032	0.003	0.001	0.027	52.991	−0.454	785
	(8.765)	(6.651)	(45.251)	(4.649)	(0.016)	(0.001)	(0.003)	(0.018)	(6.219)	(0.197)	(136)
Tamil Nadu	21.243	1.480	150.917	1.007	0.116	0.095	0.018	0.004	69.700	−0.554	1015
	(11.344)	(1.470)	(17.887)	(2.407)	(0.016)	(0.015)	(0.005)	(0.004)	(4.160)	(0.141)	(272)
Uttar	8.106	1.505	213.085	9.727	0.035	0.005	0.003	0.028	52.075	−0.477	874
Pradesh	(3.368)	(1.360)	(33.443)	(10.255)	(0.013)	(0.001)	(0.001)	(0.012)	(6.033)	(0.165)	(140)
West Bengal	34.504	3.344	159.934	7.972	0.070	0.042	0.019	0.008	66.506	−0.452	1173
	(10.718)	(1.754)	(18.859)	(11.168)	(0.015)	(0.012)	(0.004)	(0.003)	(8.728)	(0.127)	(191)
TOTAL	19.774	3.058	218.182	5.245	0.053	0.034	0.008	0.011	60.955	−0.492	1030
	(15.191)	(4.340)	(154.980)	(10.526)	(0.045)	(0.041)	(0.013)	(0.013)	(10.793)	(0.224)	(346)
Number of observations	544	539	515	527	528	524	525	524	550	552	510

Standard deviations are in parentheses. See Appendix I for details on construction and sources of variables. The data are for the sixteen main states and for the period 1958–1992. Haryana split from the state of Punjab in 1965. From this date on, we include separate observations for Punjab and Haryana. We therefore have a total of 552 possible observations. The final row gives the total number of observations available for each variable over this period.

Table I, part 2: continuation and sample counts

7.2 Figures I–III: time-series variation by state

Read:

- Figure I plots food grain production per capita by state over 1958–1992.
- Figure II plots crop flood damage per capita by state over the same period.
- Figure III plots newspaper circulation per capita by state over time.
- Food production and flood damage show large shock-like movements, while newspaper circulation tends to grow over time in many states.

Economic message: The figures illustrate both sides of the empirical design: objective needs vary sharply over time, while information capacity varies across states and grows over time.

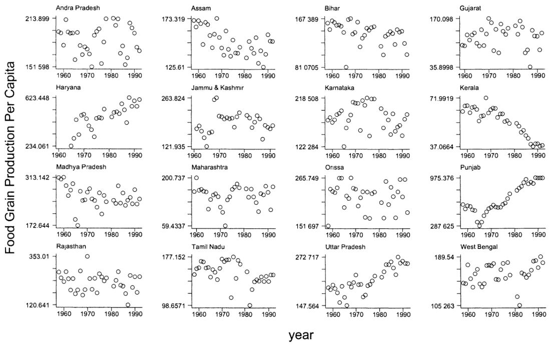

FIGURE I
Food Grain Production per Capita: 1958–1992

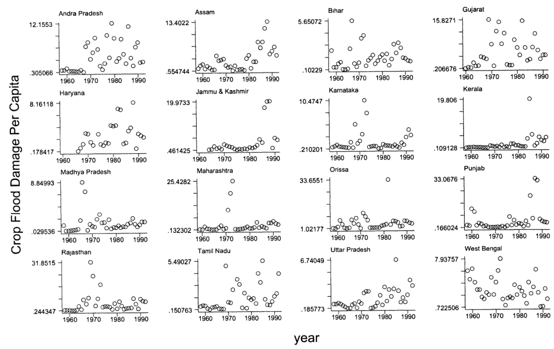

FIGURE II
Crop Flood Damage per Capita: 1958–1992

Figure I: food grain production per capita

Figure II: crop flood damage per capita

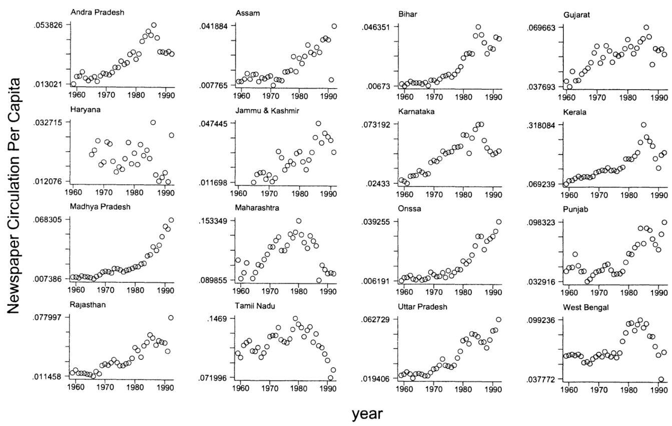

FIGURE III
Newspaper Circulation per Capita: 1958–1992

7.3 Tables II and III: shocks and government activism

Read:

- Table II shows that droughts significantly lower food grain production and that floods raise measured flood damage.
- Public food distribution rises when food grain production falls.
- Calamity relief expenditure rises when flood damage rises.
- Table III adds newspaper circulation, political variables, and economic controls.
- Newspaper circulation is positively associated with government activism in both food distribution and calamity relief.

Economic message: The public relief systems respond to objective needs, and the level of government activism is greater where accountability institutions are stronger.

TABLE II
SHOCKS AND RESPONSES IN INDIA: 1958–1992

	Food grain production	Public food distribution	Public food distribution	Flood damage	Calamity relief expenditure	Calamity relief expenditure
	(1)	(2)	(3)	(4)	(5)	(6)
Drought	−24.72			−3.510		
	(2.33)			(3.43)		
Flood	4.475			6.207		
	(0.65)			(3.20)		
Food grain production		−0.027			0.009	
		(3.55)			(1.60)	
Flood damage			0.035			0.141
			(0.79)			(4.82)
State effects	YES	YES	YES	YES	YES	YES
Year effects	YES	YES	YES	YES	YES	YES
Number of observations	460	512	524	480	507	523
Adjusted R^2	0.84	0.71	0.69	0.18	0.19	0.27

Absolute t statistics calculated using robust standard errors are reported in parentheses. See Appendix 1 for details on the construction and sources of the variables. The data are for the sixteen main states and for the period 1958–1992. Haryana split from the state of Punjab in 1965. From this date on, we include separate observations for Punjab and Haryana. We therefore have a total of 552 possible observations. Deviations from this are accounted for by missing data. Public food distribution and food grain production are expressed in per capita terms. Calamity relief expenditure and flood damage are in real per capita terms. The variables drought and flood are dummy variables for when annual average rainfall is two standard deviations below or above the state specific rainfall mean 1958–1992.

Table II: shocks and government responses

TABLE III
DETERMINANTS OF GOVERNMENT ACTIVISM

	Public food distribution			Calamity relief expenditure		
	(1)	(2)	(3)	(4)	(5)	(6)
Food grain production	−0.024	−0.026	−0.024			
	(2.51)	(2.67)	(2.43)			
Flood damage				0.149	0.146	0.144
				(4.67)	(4.72)	(4.57)
Newspaper circulation		97.19	97.82		39.84	38.63
		(3.37)	(3.60)		(2.34)	(2.25)
Turnout			−0.115			0.015
			(1.612)			(0.52)
Political competition			5.671			0.753
			(3.11)			(0.70)
Election dummy			2.497			−0.032
			(2.35)			(0.07)
Log state income	3.617	5.678	2.705	−2.258	−1.724	−2.417
	(0.69)	(1.07)	(0.51)	(0.72)	(0.54)	(0.78)
Ratio of urban to total population	130.47	71.82	62.14	−20.02	−45.54	−42.70
	(2.37)	(1.37)	(1.20)	(0.97)	(1.89)	(1.77)
Population density	−18.42	−34.03	−36.04	−9.588	−17.85	−17.29
	(0.82)	(1.76)	(1.95)	(1.56)	(2.61)	(2.59)
Log population	−43.96	−46.23	−49.59	−10.86	−9.249	−12.25
	(2.94)	(2.96)	(3.18)	(1.16)	(0.99)	(1.30)
Revenue from center	0.079	0.044	0.053	0.019	0.006	0.009
	(1.88)	(1.13)	(1.41)	(0.43)	(0.14)	(0.19)
State effects	YES	YES	YES	YES	YES	YES
Year effects	YES	YES	YES	YES	YES	YES
Number of observations	476	474	471	491	489	486
Adjusted R^2	0.75	0.76	0.77	0.27	0.28	0.28

Absolute t statistics calculated using robust standard errors are reported in parentheses. See Appendix 1 for details on the construction and sources of the variables. The data are for the sixteen main states and for the period 1958–1992. Haryana split from the state of Punjab in 1965. From this date on, we include separate observations for Punjab and Haryana. We therefore have a total of 552 possible observations. Deviations from this are accounted for by missing data. Public food distribution and food grain production are expressed in per capita terms. Calamity relief expenditure, flood damage, log state income, and revenue from center are in real per capita terms. Turnout is lagged one period and thus refers to turnout in the previous election. Political competition is defined as minus the absolute difference in the share of seats occupied by the dominant political party (Congress) and its main competitor. Election dummy captures whether it is an election or preelection year. Revenue from the

Table III: determinants of government activism

7.4 Table IV: newspapers and responsiveness

Read:

- Table IV is the core media-responsiveness table.
- Newspaper circulation interacted with food grain production is negative and significant: when production falls, public food distribution rises more in high-newspaper states.
- Newspaper circulation interacted with flood damage is positive and significant: flood damage triggers more calamity relief where newspaper circulation is higher.
- Other-language newspapers drive much of the effect, which fits the idea that local-language newspapers reach vulnerable voters more effectively.

Economic message: Media does not merely correlate with more relief. It makes relief more responsive to shocks. This is the central evidence for the accountability mechanism.

TABLE IV

Newspapers and Responsiveness

	Public food distribution				Calamity relief expenditure		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Food grain production	0.019	−0.000	−0.021	0.011			
	(0.98)	(0.00)	(2.15)	(0.56)			
Flood damage					0.063	0.144	0.085
					(2.58)	(4.46)	(2.95)
Newspaper circulation	146.84	152.34			19.41		
	(4.52)	(3.96)			(1.31)		
Newspaper circulation * food grain production	−0.444	−0.412					
	(3.11)	(2.53)					
Newspaper circulation * flood damage					1.677		
					(2.83)		
English newspaper circulation			54.64	91.63		42.97	47.76
			(0.61)	(0.68)		(0.86)	(0.96)
Hindi newspaper circulation			−14.34	−157.43		3.515	−19.33
			(0.29)	(1.18)		(0.10)	(0.52)
Other newspaper circulation			118.88	168.02		42.14	20.35
			(3.45)	(3.88)		(2.30)	(1.35)
English newspaper circulation * food grain production				−0.229			
				(0.36)			
Hindi newspaper circulation * food grain production				0.542			
				(1.09)			
Other newspaper circulation * food grain production				−0.605			
				(2.84)			
English newspaper circulation * flood damage							−5.683
							(1.70)
Hindi newspaper circulation * flood damage							2.410
							(1.29)
Other newspaper circulation * flood damage							1.964
							(3.16)
Economic controls	YES	YES	YES	YES	YES	YES	YES
Political controls	YES	YES	YES	YES	YES	YES	YES
State effects	YES	YES	YES	YES	YES	YES	YES
Year effects	YES	YES	YES	YES	YES	YES	YES
Number of observations	471	419	467	467	486	482	482
Adjusted R^2	0.77	0.76	0.77	0.77	0.30	0.28	0.30

Absolute t statistics calculated using robust standard errors are reported in parentheses. See Appendix 1 for details on the construction and sources of the variables. The data are for the sixteen main states and for the period 1958–1992. Haryana split from the state of Punjab in 1965. From this date on, we include separate observations for Punjab and Haryana. We therefore have a total of 552 possible observations. Deviations from this are accounted for by missing data. Public food distribution, food grain production, and newspaper circulation are expressed in per capita terms. Calamity relief expenditure and flood damage are in real per capita terms. "Other" captures circulation of newspapers published in languages other than English or Hindi. Food grain production in column (2) is that predicted from drought and flood variables (dummy variables for when annual average rainfall is two standard deviations below or above the state-specific rainfall mean) and state and year dummies (see column (1) of Table II). This predicted value captures the "shock" element of food production which is driven by climatic factors. Actual food grain production is used in the remainder of the regressions. The political controls are turnout lagged one period, minus the absolute difference in the share of seats occupied by the dominant political party and its main competitor and a dummy for whether it is an election or pre-election year. The economic controls are log real state income per capita, ratio of urban to total population, population density, log of total population, and revenue received from the center expressed in real per capita terms.

7.5 Table V: instrumenting newspaper circulation

Read:

- Table V instruments newspaper circulation using ownership structure.
- Ownership variables help predict newspaper circulation.
- The IV results preserve the positive role of newspapers in government activism and responsiveness.
- The paper interprets ownership structure as a proxy for media independence and transactions costs between media and government.

Economic message: The IV analysis reduces concern that states with more responsive governments simply generate more newspapers. Media structure appears to matter for policy accountability.

TABLE V

NEWSPAPERS AND RESPONSIVENESS: INSTRUMENTING WITH OWNERSHIP DATA

	Public food distribution	Public food distribution	Newspaper circulation	Calamity relief exp	Calamity relief exp	Newspaper circulation
	(1)	(2)	(3)	(4)	(5)	(6)
Food grain production	−0.023	0.055	0.000			
	(2.10)	(2.45)	(0.70)			
Flood damage				0.144	0.051	0.000
				(4.40)	(1.23)	(0.62)
Newspaper circulation	321.26	408.04		109.21	75.03	
	(2.36)	(3.14)		(2.66)	(1.87)	
Newspaper circulation * food grain production		−0.683				
		(4.73)				
Newspaper circulation * flood damage					1.758	
					(1.89)	
Share of newspapers owned by individuals			0.023			0.011
			(1.21)			(0.65)
Share of newspapers owned by public joint stock companies			−0.139			−0.127
			(1.09)			(1.05)
Share of newspapers owned by private joint stock companies			−0.028			0.002
			(0.37)			(0.03)
Share of newspapers owned by societies or associations			0.081			0.070
			(2.39)			(2.32)
Share of newspapers owned by political parties			−0.927			−0.912
			(5.19)			(5.39)

Table V, part 1: IV estimates with ownership data

Economic controls	YES	YES	YES	YES	YES	YES
Political controls	YES	YES	YES	YES	YES	YES
State effects	YES	YES	YES	YES	YES	YES
Year effects	YES	YES	YES	YES	YES	YES
Overidentification test p-value	0.97	0.91		0.97	0.98	
F-test instruments (Prob > F)			5.70			5.93
Number of observations	438	438	439	443	443	445
Adjusted R^2	0.76	0.77	0.90	0.27	0.29	0.91

Absolute t statistics calculated using robust standard errors are reported in parentheses. See Appendix 1 for details on the construction and sources of the variables. The data are for the sixteen main states and for the period 1958–1992. Haryana split from the state of Punjab in 1965. From this date on, we include separate observations for Punjab and Haryana. We therefore have a total of 552 possible observations. Deviations from this are accounted for by missing data. Public food distribution, food grain production, and newspaper circulation are expressed in per capita terms. Calamity relief expenditure and flood damage are in real per capita terms. "Other" captures circulation of newspapers published in languages other than English or Hindi. Ownership share refers to the numbers of titles under different forms of ownership expressed as share of total titles. Columns (3) and (6) present the regressions used for instrumenting newspaper circulation in columns (1), (2), and (4), (5), respectively. The overidentification test we employ is due to Sargan [1958]. The number of observations times the R^2 from the regression of the stage-two residuals on the instruments is distributed $\chi^2(T+1)$, where T is the number of instruments. The political controls are turnout lagged one period, minus the absolute difference in the share of seats occupied by the dominant political party and its main competitor and a dummy for whether it is an election or preelection year. The economic controls are log real state income per capita, ratio of urban to total population, population density, log of total population, and revenue received from the center expressed in real per capita terms.

Table V, part 2: continuation

7.6 Table VI: politics and responsiveness

Read:

- Table VI interacts shocks with turnout, political competition, and election timing.
- Newspaper circulation remains positive and significant even after political interactions are included.
- Political competition is strongly associated with public food distribution.
- Election and pre-election years are associated with higher public food distribution.
- Political interactions are weaker for calamity relief.

Economic message: Politics matters, but the pattern differs by policy. Public food distribution is more visible and politically manipulable, while calamity relief is more directly tied to disaster needs.

TABLE VI
POLITICS AND RESPONSIVENESS

	Public food distribution			Calamity relief expenditure		
	(1)	(2)	(3)	(4)	(5)	(6)
Food grain	0.041	−0.032	−0.026			
production	(0.90)	(3.13)	(3.01)			
Flood damage				−0.175	0.222	0.161
				(1.63)	(3.39)	(3.50)
Newspaper	98.73	93.55	99.49	34.97	36.07	37.95
circulation	(3.62)	(3.46)	(3.63)	(2.14)	(2.22)	(2.23)
Turnout	0.085	−0.107	−0.120	−0.018	0.012	0.015
	(0.54)	(1.51)	(1.67)	(0.66)	(0.42)	(0.53)
Turnout * food grain	−0.001					
production	(1.56)					
Turnout * flood				0.005		
damage				(2.86)		
Political competition	5.899	12.00	5.883	0.753	−0.404	0.657
	(3.20)	(3.08)	(3.21)	(0.717)	(0.32)	(0.60)
Political competition		−0.027				
* food grain production		(2.04)				
Political competition					0.182	
* flood damage					(1.69)	
Election dummy	2.535	2.420	0.061	−0.125	−0.003	0.197
	(2.36)	(2.30)	(0.03)	(0.29)	(0.01)	(0.39)
Election dummy *			0.012			
food grain production			(1.25)			
Election dummy *						−0.037
flood damage						(0.71)
Economic controls	YES	YES	YES	YES	YES	YES
State effects	YES	YES	YES	YES	YES	YES
Year effects	YES	YES	YES	YES	YES	YES
Number of observations	471	471	471	486	486	486
Adjusted R^2	0.77	0.77	0.77	0.29	0.29	0.28

Absolute t statistics calculated using robust standard errors are reported in parentheses. See Appendix 1 for details on the construction and sources of the variables. The data are for the sixteen main states and for the period 1958–1992. Haryana split from the state of Punjab in 1965. From this date on, we include separate observations for Punjab and Haryana. We therefore have a total of 552 possible observations. Deviations from this are accounted for by missing data. Public food distribution and food grain production are expressed in per capita terms. Calamity relief expenditure and flood damage are in real per capita terms. Turnout is lagged one period and thus refers to turnout in the previous election. Political competition is defined as minus the absolute value of the absolute difference in the share of seats occupied by the dominant political party and its main competitor. Election dummy is a dummy for whether it is an election or preelection year. The economic controls are log real state income per capita, ratio of urban to total population, population density, log of total population, and revenue received from the center expressed in real per capita terms.

8 Short summary

- The paper studies why governments respond differently to citizens' needs.
- The theory models responsiveness as political agency: incumbents exert effort when voters can observe and reward it.
- Media access increases the chance that vulnerable citizens learn about government action.
- Political competition and election timing increase the electoral value of responsiveness.
- The empirical setting is Indian states from 1958 to 1992.
- Need is measured using food grain production and flood damage.
- Policy responses are public food distribution and calamity relief expenditure.
- Newspaper circulation predicts higher government activism.
- Newspaper circulation also increases responsiveness to shocks.
- Local-language newspapers appear especially important.
- IV evidence using newspaper ownership supports the media-accountability interpretation.
- The paper shows that democratic accountability depends on information institutions, not just elections.

9 Conclusion

9.1 Core conclusion

Besley and Burgess show that information and political incentives shape government responsiveness. Indian state governments respond more to food production shocks and flood damage when newspaper circulation is higher and political accountability is stronger.

9.2 Economic intuition

Politicians face electoral incentives only when voters observe their actions and use that information at the ballot box. Newspapers make government actions visible to vulnerable citizens. Political competition increases the value of those voters. Together, media and competition make democratic responsiveness stronger.

9.3 Takeaway

The paper's key lesson is that democracy works through institutions that transmit information. Elections alone are not enough. Citizens also need media access and political competition for their needs to be reflected in policy.